This assignment will not be graded and is only for practice.

Level 1

1) Consider a function $f : V \times V \to \mathbb{R}$ where $V \subseteq \mathbb{R}^2$ defined by $f(v, w) = 2v_1w_1 + 5v_2w_2$, where $v = (v_1, v_2), w = (w_1, w_2)$. Choose the set of correct options. ***1 point***

- [ ] f satisfies the symmetry condition of the inner product.
- [ ] f satisfies the bilinearity condition of the inner product.
- [ ] f satisfies the positivity condition of the inner product.
- [ ] f is an inner product.
- [ ] f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f satisfies the symmetry condition of the inner product.
f satisfies the bilinearity condition of the inner product.
f satisfies the positivity condition of the inner product.
f is an inner product.

2) Consider a function $f : V \times V \to \mathbb{R}$ where $V \subseteq \mathbb{R}^2$ defined by $f(v, w) = v_1w_1 - v_1w_2 - v_2w_1 + 4v_2w_2$, where $v = (v_1, v_2), w = (w_1, w_2)$. Choose the set of correct options. ***1 point***

- [ ] f satisfies the symmetry condition of the inner product.
- [ ] f satisfies the bilinearity condition of the inner product.
- [ ] f satisfies the positivity condition of the inner product.
- [ ] f is an inner product.
- [ ] f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f satisfies the symmetry condition of the inner product.
f satisfies the bilinearity condition of the inner product.
f satisfies the positivity condition of the inner product.
f is an inner product.

3) Consider two vectors $a = (0.4, 1.3, -2.2), b = (2, 3, -5)$ in $\mathbb{R}^3$. Choose the set of correct options. **1 point**

- [ ] The two vectors satisfy the triangle inequality given by $\|a + b\| \leq \|a\| + \|b\|$.
- [ ] The two vectors do not satisfy the triangle inequality.
- [ ] The two vectors satisfy the Cauchy-Schwarz inequality given by $| < a, b > | \leq \|a\| \, \|b\|$.
- [ ] The two vectors do not satisfy the Cauchy-Schwarz inequality.

No, the answer is incorrect.
Score: 0

Accepted Answers:

The two vectors satisfy the triangle inequality given by $\|a + b\| \leq \|a\| + \|b\|$.
The two vectors satisfy the Cauchy-Schwarz inequality given by $| < a, b > | \leq \|a\| \, \|b\|$.

Level 2

4) Consider a function $f : V \times V \to \mathbb{R}$ where $V \subseteq \mathbb{R}^2$ defined by $f(v, w) = v_1^2 w_1^2 + v_1 w_2^2 + v_2^2 w_1$, where $v = (v_1, v_2), w = (w_1, w_2)$. Choose the set of correct options. **1 point**

- [ ] f satisfies the symmetry condition of the inner product.
- [ ] f satisfies the positivity condition of the inner product.

- [ ] f is an inner product.
- [ ] f is not an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f satisfies the symmetry condition of the inner product.
f is not an inner product.

5) Consider a vector $a = (a_1, b_1, c_1)$ in $\mathbb{R}^3$. Which of these is (are) possible candidates for a norm? **1 point**

- [ ] $\sqrt{a_1^2 + b_1^2 + c_1^2}$
- [ ] $a_1 - b_1$
- [ ] $a_1 + b_1 + c_1$
- [ ] $max\{a_1, b_1, c_1\}$
- [ ] $min\{a_1, b_1, c_1\}$
- [ ] $\max\{|a_1|, |b_1|, |c_1|\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\sqrt{a_1^2 + b_1^2 + c_1^2}$
$\max\{|a_1|, |b_1|, |c_1|\}$

6) Choose the set of correct statements. **1 point**

- [ ] $V = \mathbb{R}^2$ and the function $< .,. >: V \times V \longrightarrow \mathbb{R}$ defined as $< (x_1, x_2), (y_1, y_2) > = 5x_1y_1 + 8x_2y_2 - 6x_1y_2 - 6x_2y_1$ is an inner product on V.
- [ ] $V = \mathbb{R}^2$ and the function $< .,. >: V \times V \longrightarrow \mathbb{R}$ defined as $< (x_1, x_2), (y_1, y_2) > = |x_1| + 2|y_2|$ is an inner product on V.

- [ ] $V = \mathbb{R}^2$ and the function $< .,. >: V \times V \longrightarrow \mathbb{R}$ defined as $< (x_1, x_2), (y_1, y_2) >$ = $3x_1y_1 + 5x_2y_2$ is an inner product on V.
- [ ] $V = \mathbb{R}^2$ and the function $< .,. >: V \times V \longrightarrow \mathbb{R}$ defined as $< (x_1, x_2), (y_1, y_2) >$ = $3x_1y_1 - 5x_2y_2$ is an inner product on V.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V = \mathbb{R}^2$ and the function $< .,. >: V \times V \longrightarrow \mathbb{R}$ defined as $< (x_1, x_2), (y_1, y_2) >$ = $5x_1y_1 + 8x_2y_2 - 6x_1y_2 - 6x_2y_1$ is an inner product on V.
$V = \mathbb{R}^2$ and the function $< .,. >: V \times V \longrightarrow \mathbb{R}$ defined as $< (x_1, x_2), (y_1, y_2) >$ =$3x_1y_1 + 5x_2y_2$ is an inner product on V.

Your score is: 0/6